

ECET 230 - Prototype Plan - Project DeskBeam

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Purpose

This document shows planned prototyping steps for project DeskBeam.

Prototype Mauled

Goals

The goal with this prototype is to show possible investors the core features of this project and realize a proof of concept. Through this prototype we will evaluate whether some of the features suit our goal in order to choose the right combination for our end product.

Type

This is a proof-of-concept prototype.

Scope

- We will be using an esp32 as our microcontroller for this proof of concept
- Breadboard, RGB LED

Overall goal:

Ideally, we simply want to show investors that the main functions of our product work. So we want to primarily focus on the LED lighting and one of the sensors, leaning more towards the ambient light sensor.

Timeline

We are looking to push out the proof-of-concept prototype within the next two weeks.

Prototype LeBron

Goals

Prototype LeBron serves as the initial concept model for DeskBeam, establishing the foundational design direction rather than the finalized body structure. To explore this general aesthetic consensus, the initial prototyping phase will progress from low-fidelity cardboard mockups to physical wood models. Ultimately, the design will transition into SolidWorks for 3D modeling and final physical fabrication via the NJIT Makerspace 3D printers.

Type

This is a look-like prototype.

Scope

- Shows 3D-printed base
- Possibly include a display if possible (nonfunctional).
- Show the plan to integrate the remaining components in the next prototype

Timeline

We seek to finish the looks like prototype by mid-October, shortly after the initial proof-of-concept prototype.

Prototype Big Ben

Goals

This prototype demonstrates all core features in a working setup based on mainly breakout boards.

Type

This is a works-like prototype.

Scope

- We end up going with the ESP32 S3 Dev board as our microcontroller
- Breadboards/protoboards will be utilized.
- External 120 VAC to 24 VDC power adapter as our power supply
- Sensors:
    - VEML7700 digital ambient-light sensor
    - DFRobot C4001 24 GHz mmWave presence sensor
    - AHT20 temperature/humidity sensor.
    - TTP223 capacitive-touch module
    - User pushbuttons (basic tactile switches found in every electronics kit).
    - Adafruit PIR Motion Sensor
    - Touch screen display (used to display all sensor information as well as made for user interaction).
- Selecting the 24 V tunable-white COB LED strip as our light source on the lamp
- DS3231 RTC being included to implement a working clock into the design.

<u>Overview:</u>

The goal of this prototype is to have all of our electronic components up and running accordingly. This means that we will include all of the components listed on our BOM. The purpose of the works-like prototype is to give investors a better look at what the finished product will look like from a more electronics point of view.

Timeline

With gathering the necessary materials and components, we set to complete the prototype by late October.

Prototype Lionel Andrés Messi Cuccittini

Goals

This prototype will contain our end design for our study lamp while also having all the core features working as well as a more finalized chassis/body for the lamp.

Type

This is an engineering prototype.

Scope

- Will contain all final core components from the works-like prototype, and any additional features, if any, are to be added after the first prototype.
- Contains a custom PCB that will be fitted into the base of DeskBeam.
- Demonstrates use cases and performance
- Shows plan for any further ideas/changes.
- This will be close to our final version for the product pending any major changes/issues along the way of the prototyping stages.

Timeline

N/A. Depends on the progression and status of the other prototypes. However, given the course ends mid-December, we'd like to have this fully completed before then.